Mechanical Advantage Rescue Lift

Build a compact lifting mechanism that raises a simulated payload with limited input force.

Intermediate	**1-2 class periods**	**Aerospace Mission**

Engineering Context

A lunar crew must raise a damaged equipment package onto a rover platform. The crew has limited force available and needs a lightweight mechanical aid.

What You Will Practice

- Apply mechanical advantage
- Measure input and output force
- Compare ideal and actual performance
- Identify friction losses

What You Need

- String
- Pulleys or lever materials
- Spring scale
- Mass set
- Meter stick
- Engineering notebook

Student Instructions

1. Select either a pulley or lever concept.
2. Calculate the ideal mechanical advantage before building.
3. Build the mechanism with secure supports.
4. Measure the input force required to raise the assigned payload at constant speed.
5. Calculate actual mechanical advantage and efficiency.
6. Identify two locations where energy is lost.
7. Modify one feature to improve performance.
8. Repeat the force measurement and compare results.

Engineering Requirements

- Lift the assigned payload at least 20 cm
- Input force must be below the instructor-set limit
- Payload must remain controlled and not swing freely
- Include ideal MA, actual MA, and efficiency calculations

Constraints & Safety

- Use only approved masses
- Never place hands below a suspended load
- Inspect supports before every test
- Do not lift the payload above the approved height

What You Submit

- Concept sketch
- Calculations
- Force data table
- Before/after efficiency comparison
- Final recommendation

Reflection Questions

1. Where did friction reduce performance?
2. Did increasing mechanical advantage change the distance pulled?
3. What design feature improved control?
4. How would lunar gravity affect the force requirement?

Engineering Tip Move the load slowly and steadily; acceleration will distort force readings.	**Common Mistake** Ignoring the input distance can make a high mechanical advantage system seem “free.” Work must still be transferred.
Stretch Goal Add a self-locking or braking feature that holds the load without continuous input.	**Career Connection** Mechanical systems engineers design hoists, deployment mechanisms, and cargo-handling equipment for spacecraft and ground operations.